



Transport across plasma membrane:

Type I: Passive Transport

Dr. Niyan Jassim Mohammad

MBChB, MSC medical physiology

Department of medical physiology and microbiology

College of medicine

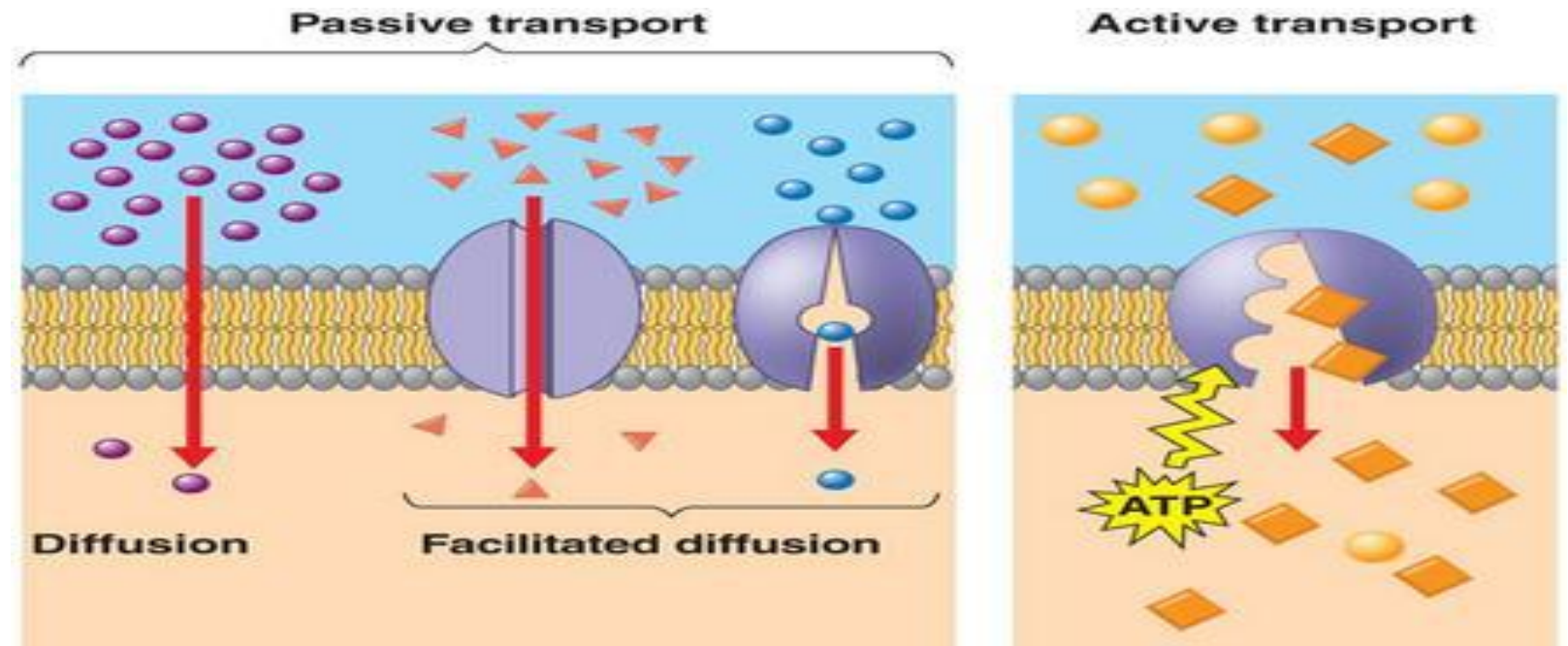
Hawler medical university

2025-2026

OBJECTIVE

- Describe the processes that transport substances across the plasma membrane.
- Identify factors that can increase the rate of diffusion.
- Describe How does simple diffusion compare with facilitated diffusion.
- Identify the osmotic pressure.
- Distinguish among isotonic, hypotonic, and hypertonic solutions.

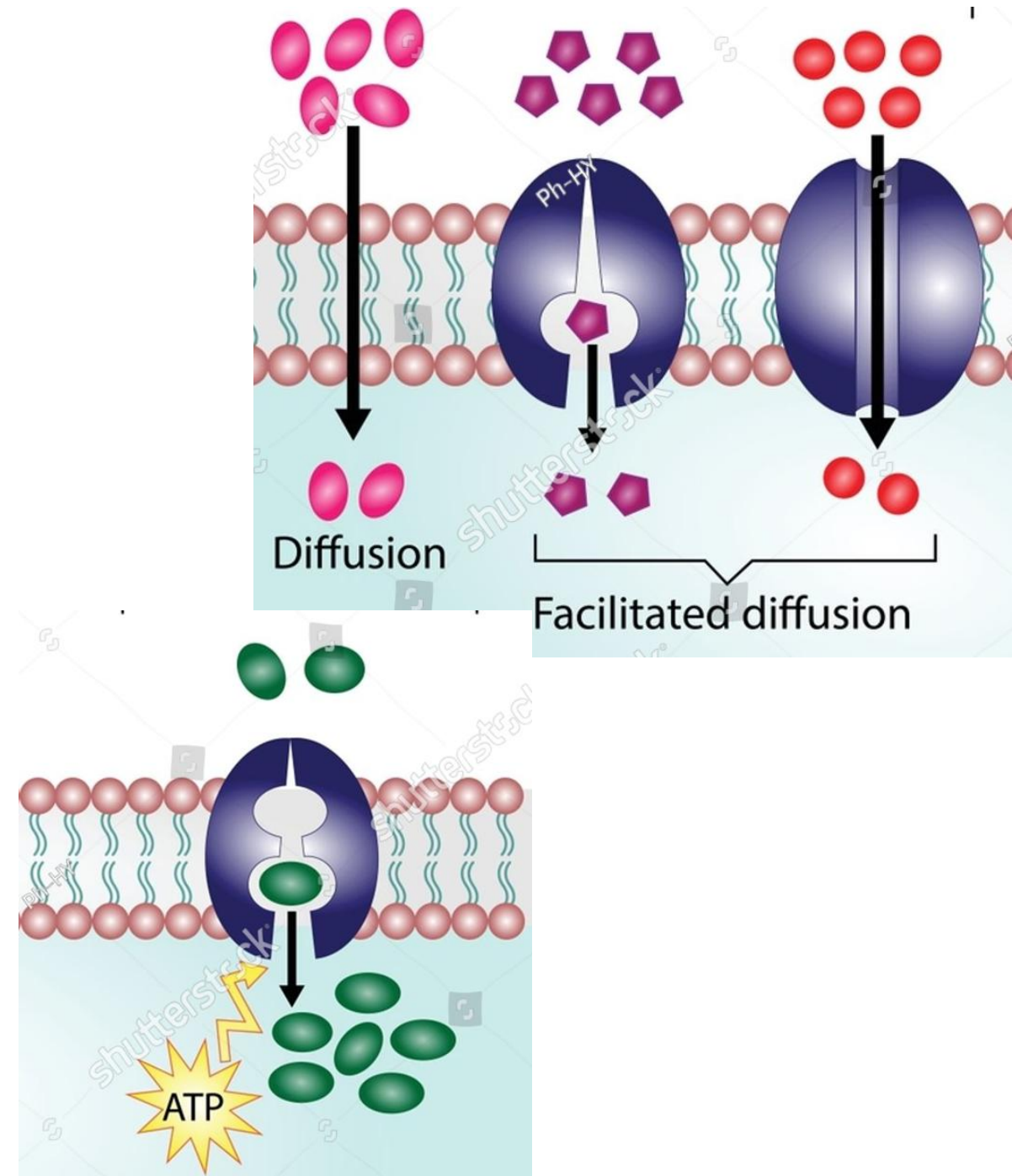
Introduction



- Transport of materials across the plasma membrane is essential to the life of a cell. Certain substances must move into the cell to support metabolic reactions. Other substances that have been produced by the cell for export or as cellular waste products must move out of the cell.
- Substances generally move across cellular membranes via transport processes that can be classified as passive or active, depending on whether they require cellular energy.

Introduction cont..

- In **passive processes**, a substance moves down its concentration or electrical gradient to cross the membrane. There is no input of energy from the cell. An example is simple diffusion.
- In **active processes**, cellular energy is used to drive the substance “uphill” against its concentration or electrical gradient. The cellular energy used is usually in the form of ATP . An example is active transport.



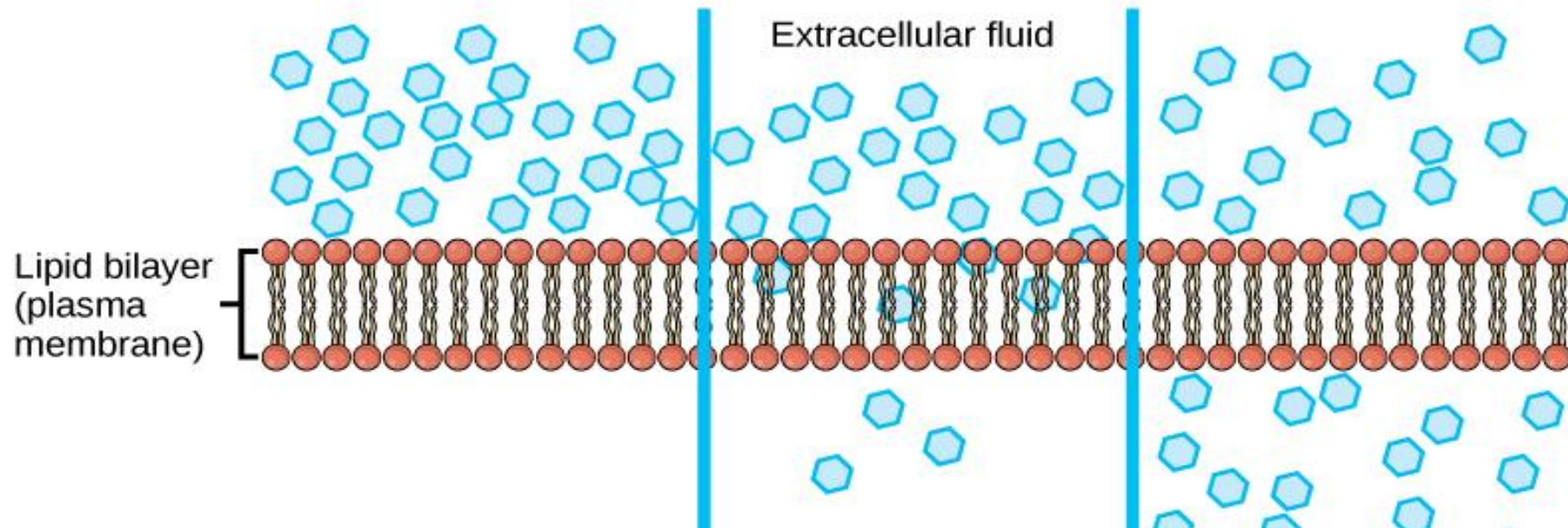
Passive Processes

The Principle of Diffusion

- **Diffusion** is a passive process in which the random mixing of particles in a solution occurs because of the particles' kinetic energy. Both the *solutes*, and the *solvent*, undergo diffusion.
- If a particular solute is present in high concentration in one area of a solution and in low concentration in another area, solute molecules will diffuse toward the area of lower concentration—they move *down their concentration gradient*.
- After some time, the particles become evenly distributed throughout the solution and the solution is said to be at equilibrium.
- The particles continue to move about randomly due to their kinetic energy, but their concentrations do not change.

factors influence the diffusion rate of substances across plasma membranes:

- **Steepness of the concentration gradient.** The greater the difference in concentration between the two sides of the membrane, the higher is the rate of diffusion. When charged particles are diffusing, the steepness of the electrochemical gradient determines the diffusion rate across the membrane.



➤ **Temperature.** The higher the temperature, the faster the rate of diffusion. All of the body's diffusion processes occur more rapidly in a person with a fever.

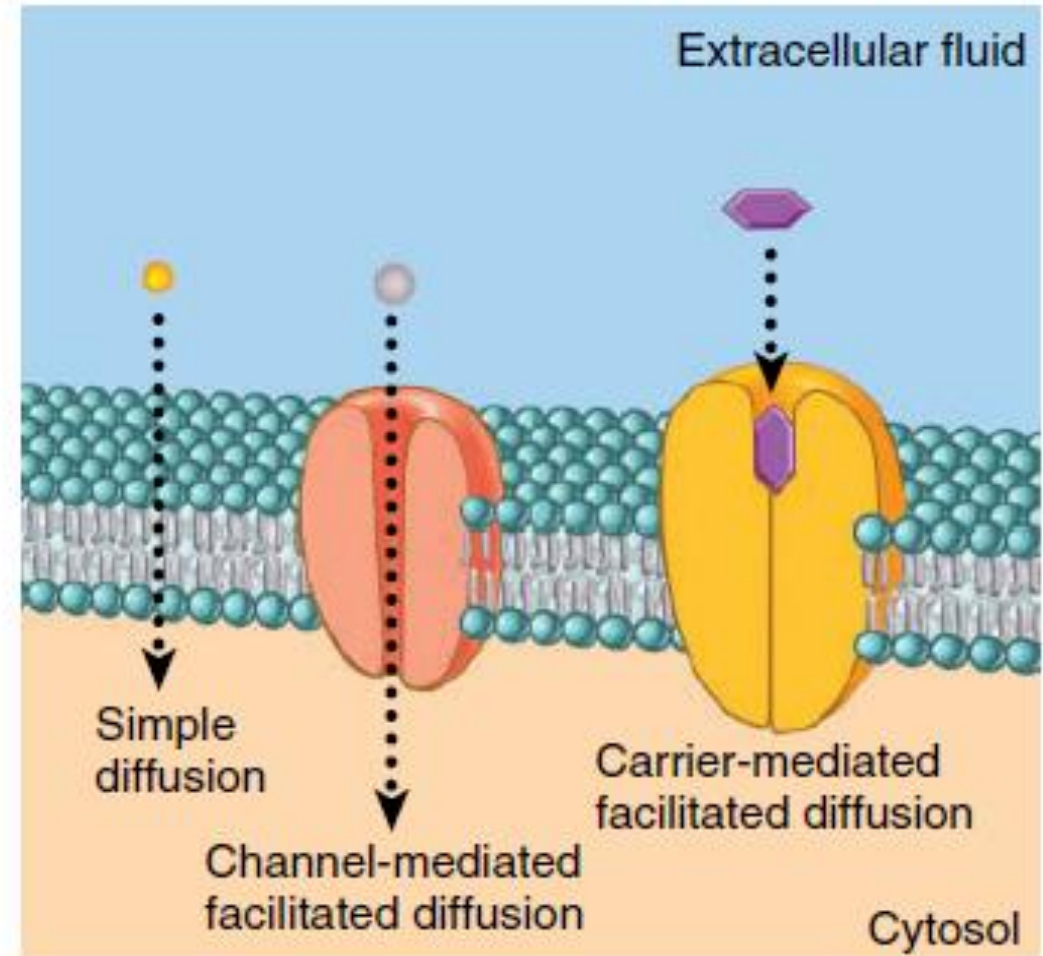
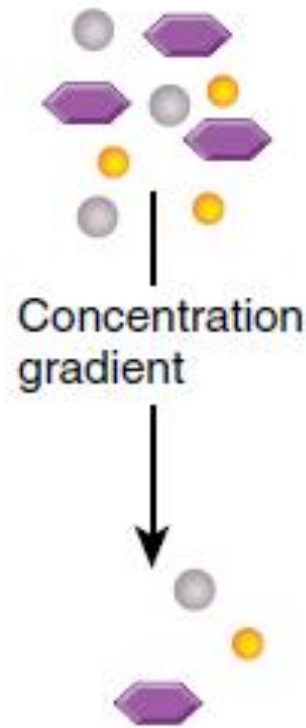


- **Mass of the diffusing substance.** The larger the mass of the diffusing particle, the slower its diffusion rate. Smaller molecules diffuse more rapidly than larger ones.
- **Surface area.** The larger the membrane surface area available for diffusion, the faster is the diffusion rate. Eg; the air sacs of the lungs have a large surface area available for diffusion of oxygen from the air into the blood. Some lung diseases, such as emphysema, reduce the surface area. This slows the rate of oxygen diffusion and makes breathing more difficult.

➤ **Diffusion distance.** The greater the distance over which diffusion must occur, the longer it takes. Diffusion across a plasma membrane takes only a fraction of a second because the membrane is so thin. In pneumonia, fluid collects in the lungs; the additional fluid increases the diffusion distance because oxygen must move through both the built-up fluid and the membrane to reach the bloodstream.

Types of diffusion

1. simple diffusion.
2. facilitated diffusion.
3. osmosis.



Simple Diffusion

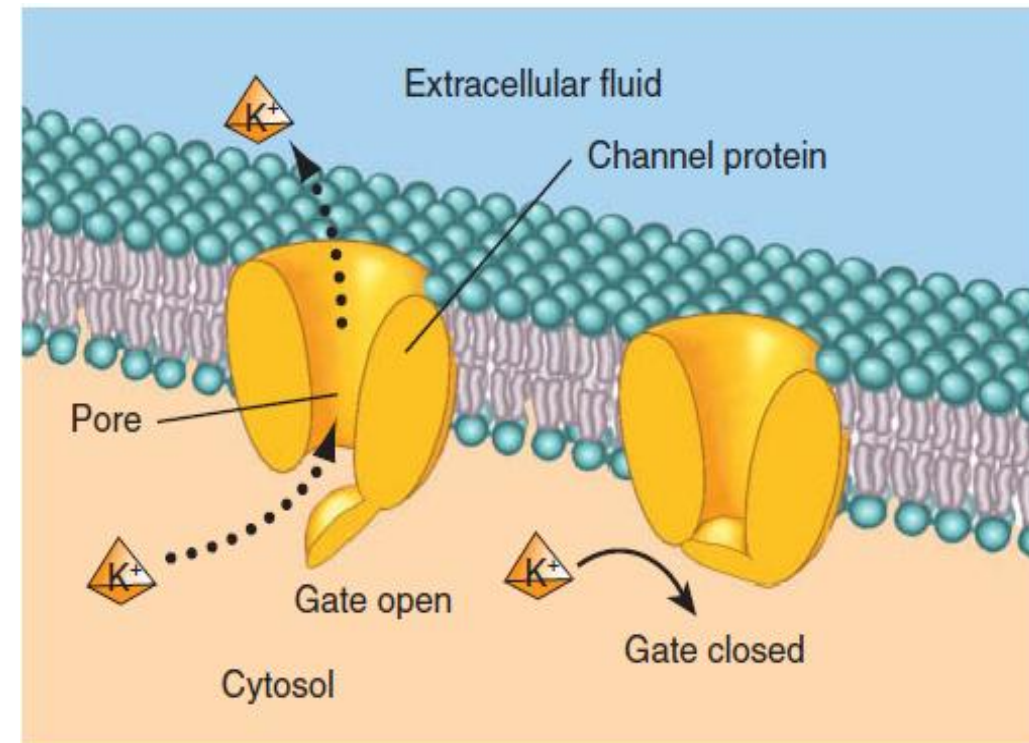
- Is a passive process
- Without the help of membrane transport proteins.
- Molecules move across the lipid bilayer through the process of simple diffusion are: Nonpolar, hydrophobic eg; oxygen, carbon dioxide, nitrogen, fatty acids, steroids, fat-soluble vitamins, water and urea.
- Simple diffusion through the lipid bilayer is important in the movement of oxygen and carbon dioxide between blood and body cells, and between blood and air within the lungs during breathing. It also is the route for absorption of some nutrients and excretion of some wastes by body cells.

Facilitated Diffusion

- Solutes that are too polar or highly charged to move through the lipid bilayer by simple diffusion can cross the plasma membrane by a passive process called **facilitated diffusion**.
- In this process, an integral membrane protein assists a specific substance across the membrane. The integral membrane protein can be either a **membrane channel or a carrier**.

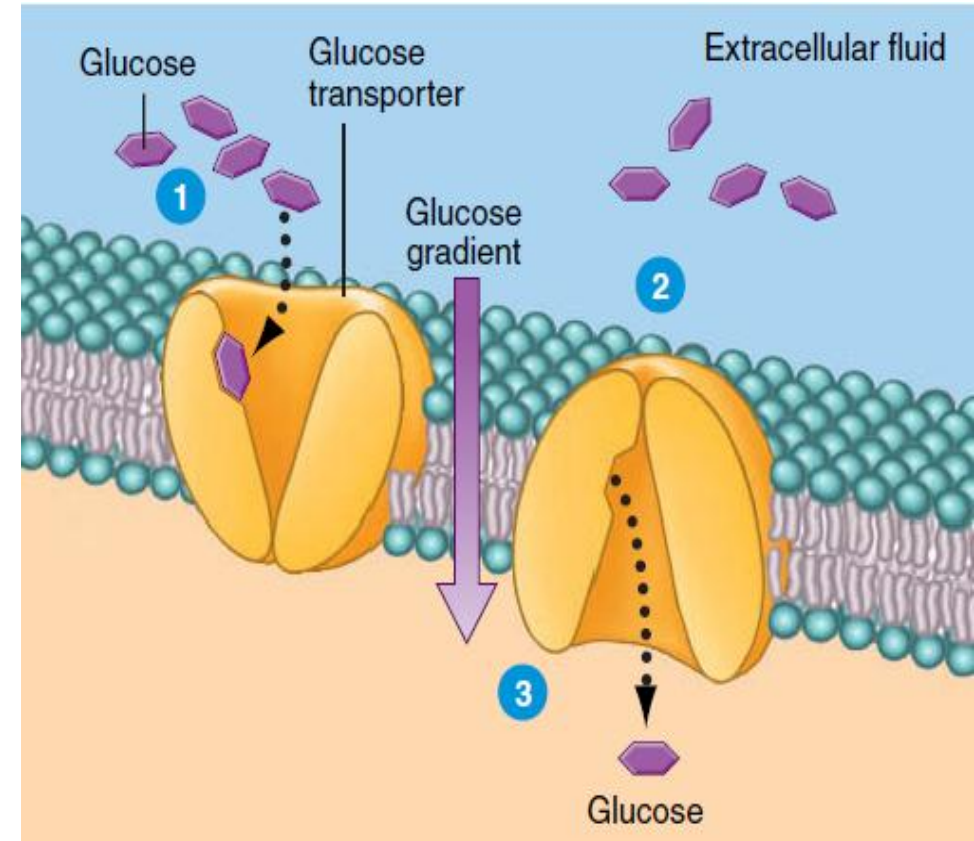
channel mediated facilitated diffusion

- solute moves down its concentration gradient through a membrane channel.
- Most membrane channels are *ion channels*; each ion can diffuse across the membrane only at certain sites.
- In typical plasma membranes, the most numerous ion channels are selective for potassium ions or chloride ions fewer channels are available for sodium ions or calcium ions.
- A channel is said to be *gated* when part of the channel protein acts as a “plug” or “gate,” changing shape in one way to open the pore and in another way to close it.



Carrier mediated facilitated diffusion:

- a *carrier* (also called a *transporter*) moves a solute down its concentration gradient across the plasma membrane.
- The solute binds to a specific carrier on one side of the membrane and is released on the other side after the carrier undergoes a change in shape.
- The solute binds more often to the carrier on the side of the membrane with a higher concentration of solute.
- Substances that move across the plasma membrane by carrier mediated facilitated diffusion include glucose, fructose, galactose, and some vitamins.



- Once the concentration is the same on both sides of the membrane, solute molecules bind to the carrier on the cytosolic side and move out to the extracellular fluid as rapidly as they bind to the carrier on the extracellular side and move into the cytosol.
- The number of carriers available in a plasma membrane places an upper limit, called the *transport maximum*, on the rate at which facilitated diffusion can occur. Once all of the carriers are occupied, the transport maximum is reached, and a further increase in the concentration gradient does not increase the rate of facilitated diffusion. Thus, much like a completely saturated sponge can absorb no more water, the process of carrier-mediated facilitated diffusion exhibits *saturation*.

Osmosis

- Is a type of diffusion in which there is net movement of a solvent through a selectively permeable membrane.
- Osmosis is a passive process. In living systems, the solvent is water, which moves by osmosis across plasma membranes from an area of higher water concentration to an area of lower water concentration.
- Another way to understand this idea is to consider the solute concentration.
- In osmosis, water moves through a selectively permeable membrane from an area of lower solute concentration to an area of higher solute concentration.

Solution's tonicity

- Is a measure of the solution's ability to change the volume of cells by altering their water content. Any solution in which a cell—for example RBC maintains its normal shape and volume is an **isotonic Solution**.
- Normally, the osmotic pressure of the cytosol is the same as the osmotic pressure of the interstitial fluid outside cells. Because the osmotic pressure on both sides of the plasma membrane (which is selectively permeable) is the same, cell volume remains relatively constant.
- When body cells are placed in a solution having a different osmotic pressure than cytosol, however, the shape and volume of the cells change. As water moves by osmosis into or out of the cells, their volume increases or decreases.

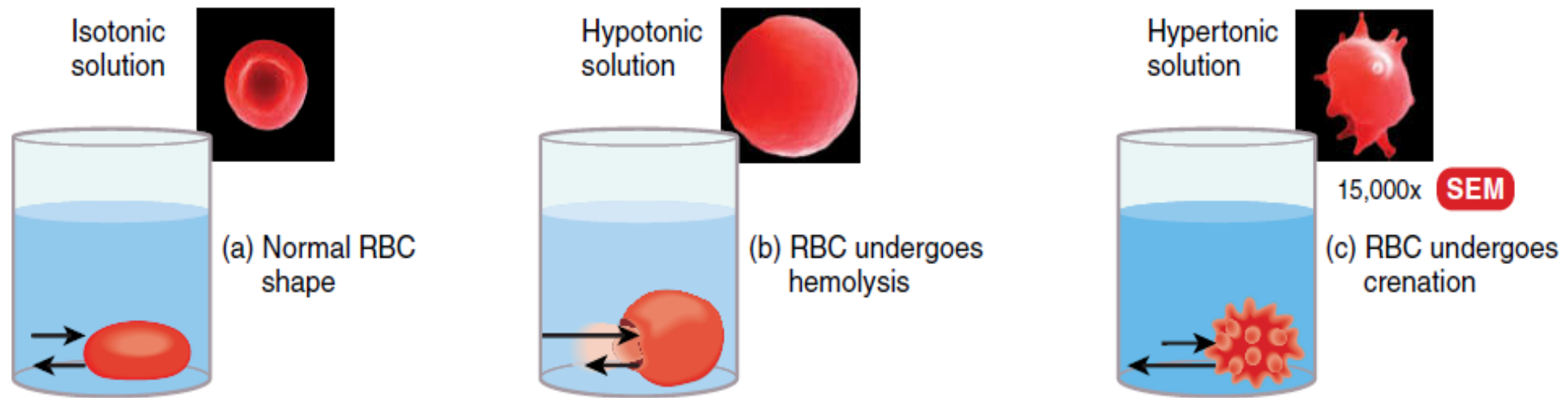
- The concentrations of solutes that cannot cross the plasma membrane are the same on both sides of the membrane in this solution. For instance, a 0.9% NaCl solution called a *normal (physiological) saline solution*, is isotonic for RBCs.
- When RBCs are bathed in 0.9% NaCl, water molecules enter and exit at the same rate, allowing the RBCs to keep their normal shape and volume.

- If RBCs are placed in a **hypotonic solution**, (a solution that has a lower concentration of solutes than the cytosol inside the RBCs). In this case, water molecules enter the cells faster than they leave, causing the RBCs to swell and eventually to burst. The rupture of RBCs in this manner is called **hemolysis**. the rupture of other types of cells due to placement in a hypotonic solution is referred to simply as **lysis**. Pure water is very hypotonic and causes rapid hemolysis.
- A **hypertonic solution** has a higher concentration of solutes than does the cytosol inside RBCs. One example of a hypertonic solution is a 2% NaCl solution. In such a solution, water molecules move out of the cells faster than they enter, causing the cells to shrink. Such shrinkage of cells is called **crenation**.

Figure 3.9 Tonicity and its effects on red blood cells (RBCs). The arrows indicate the direction and degree of water movement into and out of the cells.



 Cells placed in an isotonic solution maintain their shape because there is no net water movement into or out of the cells.





CLINICAL CONNECTION |

Medical Uses of Isotonic, Hypertonic, and Hypotonic Solutions

RBCs and other body cells may be damaged or destroyed if exposed to hypertonic or hypotonic solutions. For this reason, most **intravenous (IV) solutions**, liquids infused into the blood of a vein, are isotonic. Examples are isotonic saline (0.9% NaCl) and D5W, which stands for dextrose 5% in water. Sometimes infusion of a hypertonic solution such as mannitol (sugar alcohol) is useful to treat patients who have *cerebral edema*, excess interstitial fluid in the brain. Infusion of such a solution relieves fluid overload by causing osmosis of water from interstitial fluid into the blood. The kidneys then excrete the excess water from the blood into the urine. Hypotonic solutions, given either orally or through an IV, can be used to treat people who are dehydrated. The water in the hypotonic solution moves from the blood into interstitial fluid and then into body cells to rehydrate them. Water and most sports drinks that you consume to “rehydrate” after a workout are hypotonic relative to your body cells. •



• **THANK YOU** •

■
ANY QUESTION?